

Neural Networks Intuition

The Biological Motivation

- **Initial Goal:** Neural networks were originally invented in the 1950s with the goal of mimicking the human brain.
- **The Biological Neuron:** A biological neuron receives input signals via **dendrites**, processes them in its cell body (nucleus), and sends an output signal (an electrical impulse) down an **axon**.¹
- **Modern Context:** This analogy is now considered very loose. Our understanding of the brain is still limited, and modern neural networks are built using engineering principles, not by directly copying biology.

The Artificial Neuron

- An **artificial neuron** is a highly simplified mathematical model of its biological counterpart.²
 - **Function:**
 1. It receives one or more numerical **inputs**.
 2. It performs a simple **computation**.
 3. It produces a single numerical **output**.
 - A **neural network** is created by connecting many of these simple neurons, where the outputs from one layer of neurons become the inputs for the next layer.³
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A Brief History of Neural Networks

Neural networks have gone through several cycles of popularity and decline (sometimes called "AI winters").⁴

1. **1950s:** Initial invention.
2. **1980s–1990s:** A resurgence in popularity, showing success in applications like handwritten digit recognition (e.g., reading postal codes).⁵
3. **Late 1990s:** Fell out of favor again.
4. **~2005–Present:** A massive resurgence, rebranded as **Deep Learning**.
 - This new wave first revolutionized **speech recognition**.⁶
 - It then had a major impact on **computer vision** (e.g., the 2012 ImageNet competition).⁷
 - It subsequently transformed **Natural Language Processing (NLP)** and many other fields.⁸

Why Are Neural Networks Successful Now? 🚀

Two primary factors are responsible for the recent "deep learning revolution":

1. Massive Data Availability ("Big Data")

- Our digital society (internet, mobile phones) generates enormous amounts of data.⁹
- **Traditional algorithms** (like linear or logistic regression) **plateau** in performance. Their accuracy stops improving even when fed more data.
- **Neural networks scale with data.** Larger neural networks can effectively learn from and leverage massive datasets to achieve much higher performance.¹⁰

2. Faster Computation (Hardware)

- Training large neural networks is computationally very intensive.
- The rise of powerful **GPUs (Graphics Processing Units)**, originally built for video games, provided the perfect hardware for the parallel computations required by neural networks.¹¹